



SIMATS
ENGINEERING



SIMATS
Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

Course: Fundamentals of Data Science

Course Code: DSA0404

Slot: A

Faculty: KRISHNARAJ M

Submitted By

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Hands on Data Summarization, Data Distribution, and Outlier Treatment.

1. Complete Summary of an Employee Dataset Code:

```
File Edit Format Run Options Window Help
import pandas as pd

data = {
    "Name": ["Arun", "Ravi", "Priya", "Kiran", "Anu"],
    "Age": [25, 30, 28, 35, None],
    "Salary": [30000, 45000, 40000, 55000, 35000],
    "Department": ["IT", "HR", "IT", "Finance", "HR"]
}

df = pd.DataFrame(data)

print("Employee Dataset:")
print(df)

print("\nNumber of Rows:", df.shape[0])
print("Number of Columns:", df.shape[1])

print("\nData Types:")
print(df.dtypes)

print("\nMissing Values:")
print(df.isnull().sum())

print("\nStatistical Summary:")
print(df.describe(include="all"))
```

Output:

```
IDLE Shell 3.12.3
File Edit Shell Debug Options Window Help
Python 3.12.3 (tags/v3.12.3:f6650f9, Apr 9 2024, 14:05:25) [MSC v.1938 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
= RESTART: C:/Users/Sravan Kumar.Y/AppData/Local/Programs/Python/Python312/employeedata1.py
Employee Dataset:
   Name  Age  Salary Department
0  Arun  25.0   30000          IT
1  Ravi  30.0   45000          HR
2  Priya 28.0   40000          IT
3  Kiran 35.0   55000    Finance
4   Anu   NaN   35000          HR

Number of Rows: 5
Number of Columns: 4

Data Types:
Name          str
Age           float64
Salary        int64
Department    str
dtype: object

Missing Values:
Name          0
Age           1
Salary        0
Department    0
dtype: int64

Statistical Summary:
   Name  Age  Salary Department
count    5  4.000000  5.000000         5
unique    5      NaN      NaN         3
top  Arun      NaN      NaN         IT
freq     1      NaN      NaN         2
mean  NaN  29.500000  41000.000000      NaN
std   NaN   4.203173   9617.692031      NaN
min   NaN  25.000000  30000.000000      NaN
25%   NaN  27.250000  35000.000000      NaN
50%   NaN  29.000000  40000.000000      NaN
75%   NaN  31.250000  45000.000000      NaN
max   NaN  35.000000  55000.000000      NaN
>>>
```

2. Frequency Distribution of Students' Grades with Bar Chart

Code:

```
File Edit Format Run Options Window Help
import pandas as pd
import matplotlib.pyplot as plt

grades = ["A", "B", "A", "C", "B", "A", "B", "C", "A", "D"]
df = pd.DataFrame({"Grade": grades})

frequency = df["Grade"].value_counts().sort_index()

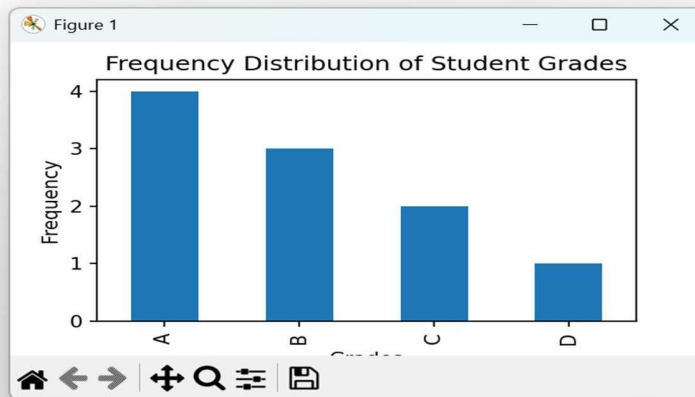
print("Frequency Distribution:")
print(frequency)

frequency.plot(kind="bar")

plt.title("Frequency Distribution of Student Grades")
plt.xlabel("Grades")
plt.ylabel("Frequency")
plt.show()
```

Output:

```
Frequency Distribution:
Grade
A      4
B      3
C      2
D      1
Name: count, dtype: int64
```



3. Histogram of House Prices and Identify Distribution Code:

houseprices.py - C:/Users/Sravan Kumar.Y/AppData/Local/Programs/Python/Python312/houseprices.py (3.12.3)

File Edit Format Run Options Window Help

```
import pandas as pd
import matplotlib.pyplot as plt

prices = [
    20, 22, 23, 24, 25, 25, 26, 27,
    28, 30, 32, 35, 40, 50, 80
]

df = pd.DataFrame({"Price": prices})

plt.hist(df["Price"], bins=6, edgecolor="black")

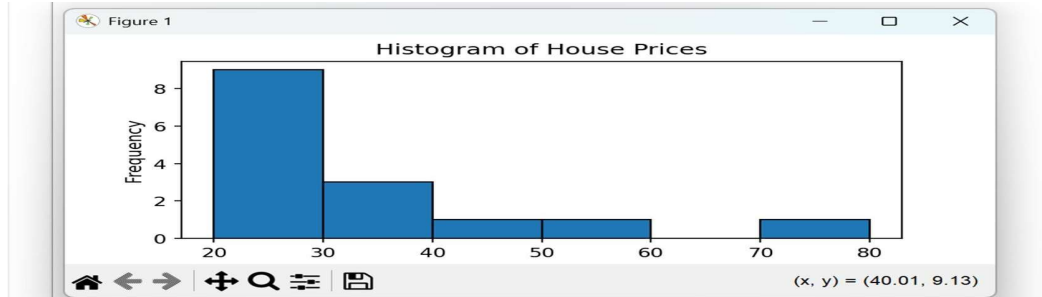
plt.title("Histogram of House Prices")
plt.xlabel("House Price")
plt.ylabel("Frequency")
plt.show()

skewness = df["Price"].skew()

print("Skewness:", skewness)

if skewness > 0.5:
    print("Data is positively (right) skewed.")
elif skewness < -0.5:
    print("Data is negatively (left) skewed.")
else:
    print("Data is approximately symmetric.")
```

Output:



4. Box Plot of Salary Data and Identify Outliers Code:

```
File Edit Format Run Options Window Help
import pandas as pd
import matplotlib.pyplot as plt

salary = [
    25000, 28000, 30000, 32000, 35000,
    36000, 38000, 40000, 42000, 100000
]

df = pd.DataFrame({"Salary": salary})

plt.boxplot(df["Salary"])

plt.title("Box Plot of Salary")
plt.ylabel("Salary")
plt.show()

Q1 = df["Salary"].quantile(0.25)
Q3 = df["Salary"].quantile(0.75)

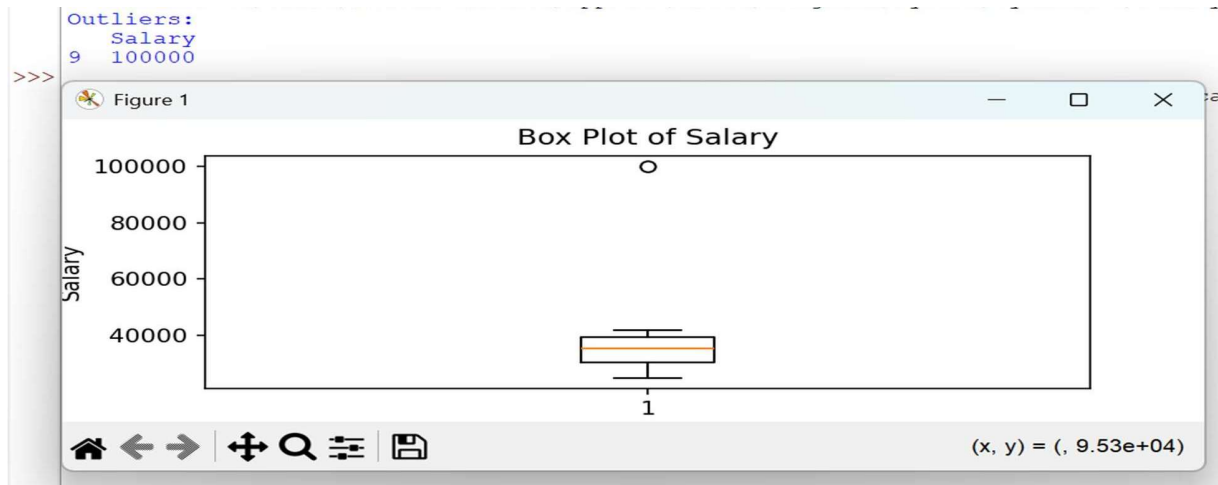
IQR = Q3 - Q1

lower = Q1 - 1.5 * IQR
upper = Q3 + 1.5 * IQR

outliers = df[
    (df["Salary"] < lower) |
    (df["Salary"] > upper)
]

print("Outliers:")
print(outliers)
```

Output:



5. Detect Outliers in Students' Marks Using IQR Code:

```
File Edit Format Run Options Window Help
import pandas as pd

marks = [45, 50, 52, 55, 58, 60, 62, 65, 68, 70, 95, 10]

df = pd.DataFrame({"Marks": marks})

Q1 = df["Marks"].quantile(0.25)
Q3 = df["Marks"].quantile(0.75)

IQR = Q3 - Q1

lower = Q1 - 1.5 * IQR
upper = Q3 + 1.5 * IQR

outliers = df[
    (df["Marks"] < lower) |
    (df["Marks"] > upper)
]

print("Q1:", Q1)
print("Q3:", Q3)
print("IQR:", IQR)

print("\nOutliers:")
print(outliers)
```

IDLE Shell 3.12.3

```
File Edit Shell Debug Options Window Help
Python 3.12.3 (tags/v3.12.3:f6650f9
AMD64)] on win32
Type "help", "copyright", "credits"
>>>
= RESTART: C:/Users/Sravan Kumar.Y/i
IQR.py
Q1: 51.5
Q3: 65.75
IQR: 14.25

Outliers:
  Marks
10    95
11    10
>>>
```

6. Remove Outliers from a Sales Dataset Using IQR

Code:

```

File Edit Format Run Options Window Help
import pandas as pd

sales = [100, 120, 125, 130, 140, 150, 160, 170, 180, 1000]
df = pd.DataFrame({"Sales": sales})

print("Original Dataset:")
print(df)

Q1 = df["Sales"].quantile(0.25)
Q3 = df["Sales"].quantile(0.75)

IQR = Q3 - Q1

lower = Q1 - 1.5 * IQR
upper = Q3 + 1.5 * IQR

cleaned_df = df[
    (df["Sales"] >= lower) &
    (df["Sales"] <= upper)
]

print("\nCleaned Dataset:")
print(cleaned_df)

```

```

IDLE Shell 3.12.3
File Edit Shell Debug Optic
Python 3.12.3 (tags/
AMD64) on win32
Type "help", "copyri
>>>
= RESTART: C:/Users/
ingoutliers.py
Original Dataset:
  Sales
0    100
1    120
2    125
3    130
4    140
5    150
6    160
7    170
8    180
9   1000

Cleaned Dataset:
  Sales
0    100
1    120
2    125
3    130
4    140
5    150
6    160
7    170
8    180
>>>

```

7. Histograms and Box Plots Before and After Removing Outliers Code:

```

File Edit Format Run Options Window Help
import pandas as pd
import matplotlib.pyplot as plt

data = [10, 12, 13, 14, 15, 16, 17, 18, 19, 20, 100]
df = pd.DataFrame({"Values": data})

# Before removing outliers
plt.hist(df["Values"], bins=6, edgecolor="black")
plt.title("Histogram Before Removing Outliers")
plt.show()

plt.boxplot(df["Values"])
plt.title("Box Plot Before Removing Outliers")
plt.show()

# Calculate IQR
Q1 = df["Values"].quantile(0.25)
Q3 = df["Values"].quantile(0.75)

IQR = Q3 - Q1

lower = Q1 - 1.5 * IQR
upper = Q3 + 1.5 * IQR

# Remove outliers
cleaned_df = df[
    (df["Values"] >= lower) &
    (df["Values"] <= upper)
]

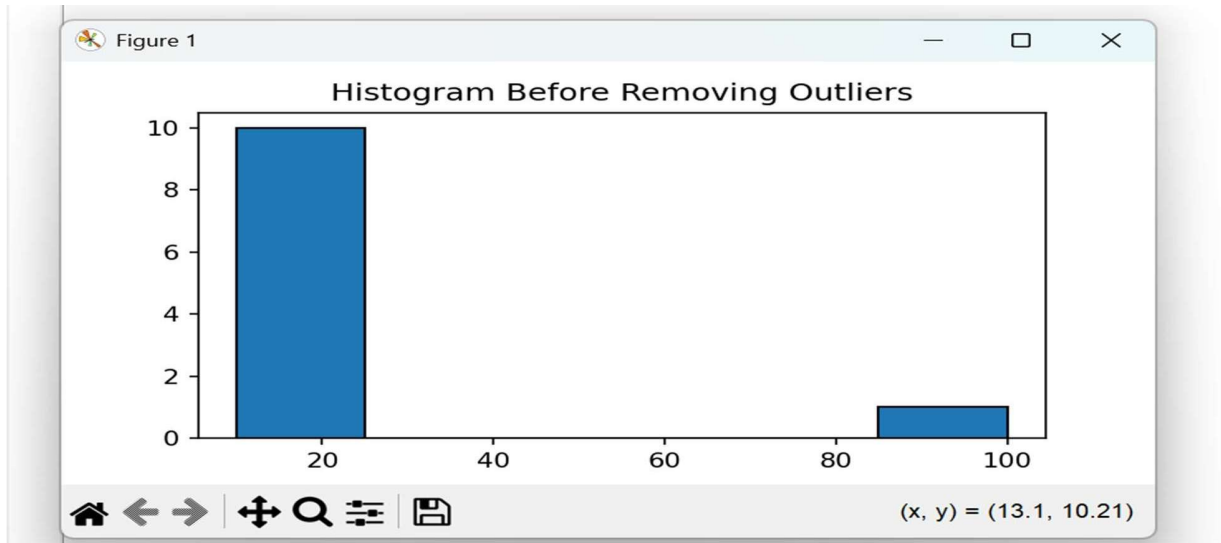
# After removing outliers
plt.hist(cleaned_df["Values"], bins=6, edgecolor="black")
plt.title("Histogram After Removing Outliers")
plt.show()

plt.boxplot(cleaned_df["Values"])
plt.title("Box Plot After Removing Outliers")
plt.show()

print("Cleaned Dataset:")
print(cleaned_df)

```

Output:



8. EDA on Iris Dataset

Code:

eda.py - C:/Users/Sravan Kumar.Y/AppData/Local/Programs/Python/Python312/eda.py (3.12.3)

```
File Edit Format Run Options Window Help
plt.suptitle("Histograms of Iris Dataset")
plt.show()

# Box plots
df.boxplot()
plt.title("Box Plots of Iris Dataset")
plt.xticks(rotation=20)
plt.show()

# Detect and remove outliers using IQR
numeric_columns = iris.feature_names

cleaned_df = df.copy()

for column in numeric_columns:

    Q1 = cleaned_df[column].quantile(0.25)
    Q3 = cleaned_df[column].quantile(0.75)

    IQR = Q3 - Q1

    lower = Q1 - 1.5 * IQR
    upper = Q3 + 1.5 * IQR

    outliers = cleaned_df[
        (cleaned_df[column] < lower) |
        (cleaned_df[column] > upper)
    ]

    print("\nOutliers in", column)
    print(outliers)

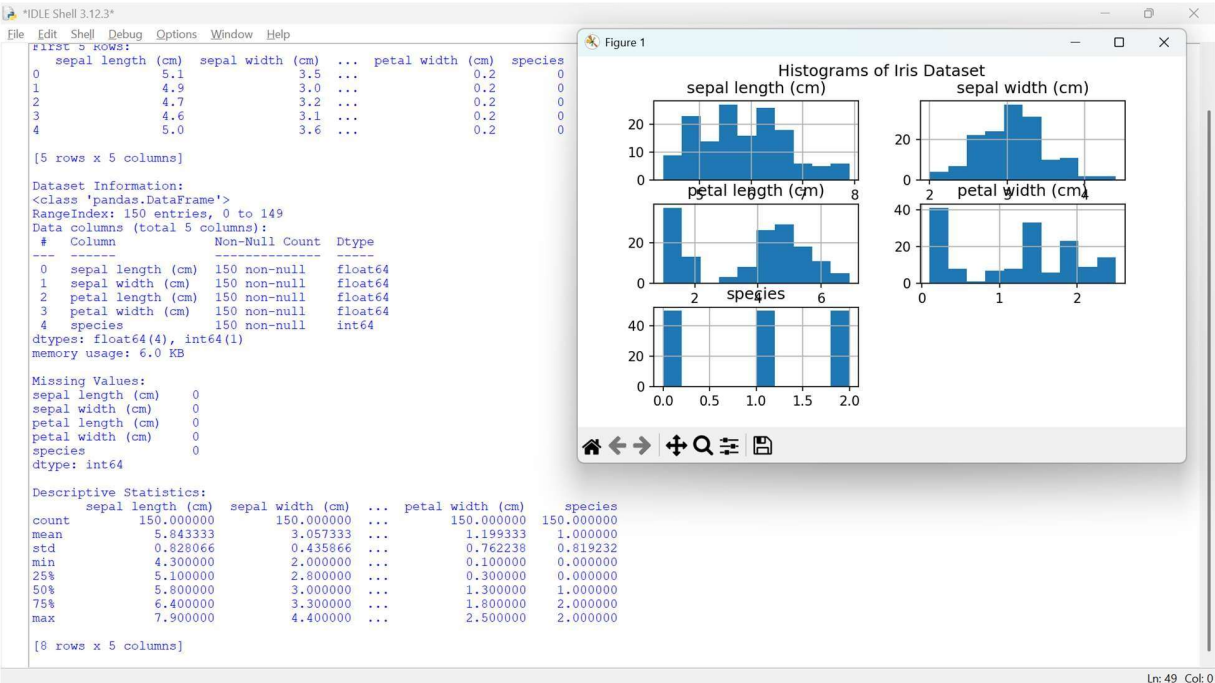
    cleaned_df = cleaned_df[
        (cleaned_df[column] >= lower) &
        (cleaned_df[column] <= upper)
    ]

# Save cleaned dataset
cleaned_df.to_csv("cleaned_iris.csv", index=False)

print("\nOriginal Dataset Size:", df.shape)
print("Cleaned Dataset Size:", cleaned_df.shape)

print("\nCleaned dataset saved as cleaned_iris.csv")
```

Output:



```
IDLE Shell 3.12.3*
File Edit Shell Debug Options Window Help

First 5 rows:
  sepal length (cm)  sepal width (cm)  ...  petal width (cm)  species
0                5.1                3.5  ...              0.2         0
1                4.9                3.0  ...              0.2         0
2                4.7                3.2  ...              0.2         0
3                4.6                3.1  ...              0.2         0
4                5.0                3.6  ...              0.2         0

[5 rows x 5 columns]

Dataset Information:
<class 'pandas.DataFrame'>
RangeIndex: 150 entries, 0 to 149
Data columns (total 5 columns):
#   Column                Non-Null Count  Dtype
---  ---                ---
0   sepal length (cm)      150 non-null   float64
1   sepal width (cm)       150 non-null   float64
2   petal length (cm)      150 non-null   float64
3   petal width (cm)       150 non-null   float64
4   species                150 non-null   int64
dtypes: float64(4), int64(1)
memory usage: 6.0 KB

Missing Values:
sepal length (cm)    0
sepal width (cm)     0
petal length (cm)    0
petal width (cm)     0
species              0
dtype: int64

Descriptive Statistics:
   count  sepal length (cm)  sepal width (cm)  ...  petal width (cm)  species
count    150.000000      150.000000  ...    150.000000    150.000000
mean       5.843333        3.057333  ...     1.199333     1.000000
std        0.828066        0.435866  ...     0.762238     0.819232
min         4.300000        2.000000  ...     0.100000     0.000000
25%         5.100000        2.800000  ...     0.300000     0.000000
50%         5.800000        3.000000  ...     1.300000     1.000000
75%         6.400000        3.300000  ...     1.800000     2.000000
max         7.900000        4.400000  ...     2.500000     2.000000

[8 rows x 5 columns]
```